

Lecture 8

Particles and Waves

Art and Lenny have had enough entanglement for now. They're ready for something simpler.

Lenny: *Hey Hilbert, do you have anything in one dimension?*

Hilbert: *Let me check. Single dimensions are very popular lately. Sometimes we run out.*

Art: I'd settle for something classical, if that's all you have.

Hilbert: *Not here, friend. We'd lose our license*.

Art: *Good point.*

To the person in the street, quantum mechanics is all about light being particles and electrons being waves. But up until now, I've hardly mentioned particles, and the only mention of waves has been the wave function, which so far has had nothing to do with waves. So when do we get to the "real" quantum mechanics?

The answer, of course, is that real quantum mechanics is not so much about particles and waves as it is about the nonclassical logical principles